

T4 vs CPU-Only Cost Analysis

1. Modal GPU Tiers (Below T4)

There is NO GPU tier below T4 on Modal. The only option below T4 is CPU-only. Modal does not offer older GPUs like K80 or P100.

2. Modal Pricing (2026)

Resource	Per Second	Per Hour	Notes	Verdict
T4 GPU	\$0.000164	\$0.59	16GB VRAM	Cheapest GPU
CPU (4 cores)	\$0.000158	\$0.57	2 vCPU = 1 core	Similar cost/hr
Memory (16GB)	\$0.000036	\$0.13	GiB/sec	Additional cost

3. Per-Call Cost Comparison

The critical metric is cost PER INFERENCE CALL, not cost per hour. CPU is much slower, so each call takes longer and costs more.

Resource	Time/Call	Cost/Call	Monthly (100/day)	Winner
T4 GPU	8 sec	\$0.0013	\$3.90	T4
CPU 4-core	180 sec	\$0.028	\$8.40	

T4 is 21x CHEAPER per call than CPU-only.

4. Why CPU-Only Is Worse

CPU inference for LLMs is extremely slow because CPUs lack the parallel processing power that GPUs provide. Llama-3-8B in 4-bit needs 10-12GB of matrix operations per inference, which GPUs handle in parallel but CPUs process sequentially.

Factor	T4 GPU	CPU 4-core
Inference time	4-8 seconds	2-5 minutes
Cost per call	\$0.0013	\$0.028
Monthly (100/day)	\$3.90	\$8.40
Annual cost	\$46.80	\$100.80
User experience	Instant	Slow

5. Summary

T4 is the cheapest and fastest option for LLM inference on Modal.

Below T4, there is no cheaper GPU. CPU-only is available but costs MORE per call because it is 22x slower. T4 at \$0.59/hr is the optimal choice for cost-sensitive deployments.